Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · UE26ME142A Unit 3 Class 44.1 AV Summary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 3 Class 44.1 AV Summary

1 UE26ME142A Unit 3 Class 44.1 AV Summary

Study focus: UE26ME142A_Unit 3_Class 44.1_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000186 · EV0000187

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Venkitesh, a civil engineer from the Department of Science and Humanities, PS University.
- This is class number 56.
- We'll be continuing to solve few more problems in today's class.
- And we are discussing about the.
- Unit 3 distributed forces in that chapter 5, beams internal effect, and we are referring to the engineer mechanics statics book written by Miriam and Craig, and it is the 8th edition.
- The problem number is 5 bar 130.
- The statement of the problem goes like this.
- Determine the shear force V and bending moment M at a section of the loaded beam 200 mm to the right of A.
- It's a simply supported beam.
- At the left hand, that is at the point A, you have a hint support or a pin connection or pin support, and then at the right end, that is at point B, we have a roller support.
- And to this beam, there is an uniformly distributed load or a rectangle loading of intensity 6 kN per meter for a distance of 300 mm from the left support that is from A.
- The length of the load is three hundred mm, whereas the total length of the beam is three hundred mm plus three hundred mm, totally it is six hundred mm.
- 200 mm will be somewhere here.
- So if I want to determine the shear force and bending moment, first thing that I need to do is to find out what is the support reactions.

- To determine the support reactions.
- I have to draw the free body diagram.
- To draw the free body diagram I have to replace the support by its respective supports.
- That is one vertical reaction at A and one horizontal reaction, vertical reaction BA and horizontal reaction H A.
- Then at B I will replace this by one reaction that is R B which is acting vertically upwards.
- Then the uniformly distributed load will be replaced by a point load of magnitude equal to the.
- So once I draw this free body diagram, next I'll apply the equations of equilibrium, all the three equations of equilibrium, to determine the vertical reaction at A, horizontal reaction at A, and then vertical reaction at B.
- Then reaction at B by R B and then the UDL by a point load of magnitude 1.8 kilonewtons, which is acting at a distance of 0.15 meters from A.
- I get V A into this total distance, which is 0.6 meters, which will produce a clockwise movement.
- Whereas this one would produce an anti-clockwise moment, so it is minus 1.8 into the distance from this to this is 0.6 minus 0.15, which is 0.14, that's what is mentioned here.

Example or application from the lecture

- And the diagram that is given here is the diagram that is given.
- Length of the load that is 0.3 mm because the intensity of the loading is given as 6 kN per meter to have consistent in units, I'll convert this distance 300 mm into 0.3 meters.
- Once I get these reactions, then I can consider the section at 200 mm from a I can take either left side of the section or the light side of the section.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 05:00–09:59

We will get V A to be equal to 1.35 kilon and we got a positive value.

Key details

- So that means the sense of VA is acting upwards.
- Next, once I get this, of course I can determine the reaction at B by using.
- The algebraic sum of all the vertical forces acting to be equal to zero.
- But however, since I have to determine the shear force and bearing moment at 200 mm from A, I need not.
- I have to determine RB.
- Instead I can take the section of the beam to the left side and then solve it easily.
- So that's why I have not determined the reaction at B or the reaction at HA of obviously H A is going to be zero because there is no horizontal force that is acting.
- The section 200 mm.
- What do we have?
- We have VA which is acting vertically upwards.

- Then very importantly, what we need to draw here is whatever is the load that is acting till 200 mm is what we need to take.
- UDL which is acting of my intensity 6 kN per meter and for a distance of 0.2 meters.
- The point load was 6 into 0.3.
- It's the total load that we are taking, whatever is the load that is acting on the beam.
- But when we are taking a section to find out the shear force embedding moment, which is at point two meters from A, we have to take only that load.
- I hope this is very clear.
- The force is acting upward that is V A, so it is positive, which has a magnitude of one point three five kilonewton.
- Then to the left of the section and downwards, so it would be negative of the value 1.2, so that is what is mentioned here.
- That is the shear force that is acting at the section 200 mm to the right of A.
- Similarly to determine the bending movement, so bending movement at this point will be equal to V VA into point two, which is the moment, and that causes a sagging bending movement because the beam is going to bend like this.
- Then because of this load, the beam is going to bend like this, which is a hogging bending movement.
- The distance 0.1, which is mentioned here.
- support A and that's what we have done.
- You can see the free body diagram of the beam.
- The UDL has been converted into a point load and represented as R.
- Taking moment about B we get RA, which is a value 1.35 kilonewton.
- Where we want the shear force of minimum that is 200 mm from A.
- This is the diagram.
- The sum of all the forces acting unequality to zero, so you get the shear V is equal to 0.15 kilonaut time, then takes algebraic sum of all the movements acting at A to be equal to 0.

Example or application from the lecture

- Earlier when we consider the entire beam.
- What is the value of R is given here, that is 6 into 0.3, which is 1.8 kilonewton.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 10:00–14:59

But the section 200 mm from A, which is equal to 0.15 kilonewton per meter.

Key details

- I hope all of you have understood this problem.
- So next we will move on and this is the next problem that I have considered.
- Problem number 5 bar 132.
- The problem statement goes like this.

- Determine the shear force V and bending moment M in the beam at a section 2 meter to the right of end A.
- The beam was subjected to a UDL, whereas in this problem, the beam is a simply supported beam again, of span 6 meters.
- The left support is a hinge support, the right support is a roller support.
- But here the loading is a uniformly varying load or a triangular load with zero intensity at point A and maximum intensity of 4.8 kN per meter at a distance of 3 meters from A.
- Once you get the reaction at A.
- Draw a separate diagram.
- I have drawn the free body diagram of the beam, replacing the support by its reaction B A and H A at point A where there was hinge.
- Then the triangular loading.
- The magnitude of that would be half of 3, which is the span of the loading into the intensity of loading, which is 4.8.
- The point of application of this is going to be at a distance of two thirds from A.
- Then at point P the roller support is replaced by a reaction R B, vertical reaction R B.
- Four point eight kilonewtons what we have got.
- Next we will draw only the.
- Forces that is acting at 2 meters from V A.
- Considering the left side of the support, so this is what we get.
- We get a vertical reaction VA, which is of 4.8 kilonewtons.
- For 2 meters a triangular loading.
- When we have to convert this triangular loading into a point load, we should find out what is the intensity of the load at two meters from A.
- This is one triangle, and then this is another triangle, they are similar.
- of 2 from A, which I have mentioned it here.
- 2 thirds of 2 that is 4 by 3.
- But section 2 meters from the right of End A, shear force will be we have VA to the left of the section and is acting upwards, so it is positive, whereas 3.2 is to the left and downwards, it is negative.
- Coming to bending moment, considering the moment at two meters from point A.
- We have VA into A which will cause a saging bending moment, so it's positive, whereas 3.2 will cause an hogging bending movement, so it is negative, so it's 3.2 into the distances 2 by 3.

Example or application from the lecture

- Consider a section 2 meters from A.
- Consider all the forces that is acting in this 2 meter span of the beam, and then find out the shear force and bending movement at that point.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 15:00–19:59

Sorry, I have left out the kilonewton meters here.

Key details

- Please kindly make that correction.
- Replacing the triangular loading by a point load, then determining what is the reaction at RA, which is 4.8 kilon, then considering only the portion of the beam to the left side of the section that is at 2 meters from here.
- Whatever is the loading that is acting within this pan, that is for two meters, this is a triangle.
- We have EA acting here.
- Then determining the distance at which this R1 is acting, which he has considered as A.
- Which is 2 thirds of 2, which is 4 by 3 meters.
- Then R1 is half of 3.2 into 2, that is 3.2 kilonewtons.
- Using this, we can find out the shear force considering all the vertical forces acting on the beam.
- Then taking movement about this point and equating it to 0, we get.
- Minus 3.2 into 4 by 3 which is this force.
- Then minus one point six into two.
- We are equal to zero.
- We are equal to zero.
- I hope everybody is clear with this.
- If you have any doubts, you can always contact me.
- I am Dr.
- S V Vength, Civil Engineering from Department of Science and Humanities, PS University.
- See you in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 3_Class 44.1_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/044_UE26ME142A_Unit_3_Class_44.1_AV_Summary__02d4870a880196ad.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.

4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

C Revision and retrieval practice